

Take Home Activity 05

Student Number – 35040

Q1

```
1 public class HotelRoom {
2     int roomNumber;
3     String roomType;
4     double pricePerNight;
5     boolean isBooked;
6
7     HotelRoom() {
8         roomNumber = 0;
9         roomType = "Standard";
10        pricePerNight = 5000.0;
11        isBooked = false;
12    }
13
14    HotelRoom(int roomNumber, String roomType, double pricePerNight, boolean isBooked) {
15        this.roomNumber = roomNumber;
16        this.roomType = roomType;
17        this.pricePerNight = pricePerNight;
18        this.isBooked = isBooked;
19    }
20
21    double bookRoom(int nights) {
22        if (!isBooked) {
23            isBooked = true;
24            return pricePerNight * nights;
25        } else {
26            System.out.println("Room is already booked.");
27            return 0.0;
28        }
29    }
30
31    void checkoutRoom() {
32        isBooked = false;
33        System.out.println("Room " + roomNumber + " is now available.");
34    }
35
36    String getRoomSummary() {
37        return "Room " + roomNumber + " - " + roomType + " - Rs." + pricePerNight + " per night - Booked:" + isBooked;
38    }
39
40    public static void main(String[] args) {
41        HotelRoom room1 = new HotelRoom();
42        HotelRoom room2 = new HotelRoom(101, "Deluxe", 8500.0, false);
43
44        double cost = room1.bookRoom(3);
45        System.out.println("Total cost: " + cost);
46
47        room1.bookRoom(2);
48
49        room1.checkoutRoom();
50
51        double newCost = room1.bookRoom(2);
52        System.out.println("New total cost: " + newCost);
53
54        System.out.println(room1.getRoomSummary());
55        System.out.println(room2.getRoomSummary());
56    }
57 }
58
59
60
61
62
63
64
65
66
```

```
Total cost: 15000.0  
Room is already booked.  
Room0 is now available.  
New total cost: 10000.0  
Room0-Standard- Rs.5000.0per night - Booked:true  
Room101-Deluxe- Rs.8500.0per night - Booked:false
```

Q2

```
1 public class OnlineOrder {
2     int orderId;
3     String customerName;
4     double itemPrice;
5     int quantity;
6
7     public OnlineOrder(int orderId, String customerName, double itemPrice, int quantity) {
8         this.orderId = orderId;
9         this.customerName = customerName;
10        this.itemPrice = itemPrice;
11        this.quantity = quantity;
12    }
13
14    public double calculateTotal(double discountPercentage) {
15        double total = itemPrice * quantity;
16        double discount = total * discountPercentage / 100;
17        return total - discount;
18    }
19
20    public void updateQuantity(int newQuantity) {
21        if (newQuantity > 0) {
22            quantity = newQuantity;
23            System.out.println("Quantity updated to " + newQuantity + ".");
24        } else {
25            System.out.println("Invalid quantity.");
26        }
27    }
28
29    public String getOrderSummary() {
30        return "Order ID: " + orderId +
31            ", Customer Name: " + customerName +
32            ", Item Price: Rs. " + itemPrice +
33            ", Quantity: " + quantity;
34    }
35
36    public static void main(String[] args) {
37        OnlineOrder order1 = new OnlineOrder(1, "Nimal", 2500.0, 2);
38        OnlineOrder order2 = new OnlineOrder(2, "Kamal", 1500.0, 3);
39
40        double finalAmount = order1.calculateTotal(10);
41        System.out.println("Final amount after 10% discount: Rs. " + finalAmount);
42
43        order1.updateQuantity(5);
44        order1.updateQuantity(-2);
45
46        System.out.println(order1.getOrderSummary());
47        System.out.println(order2.getOrderSummary());
48    }
49 }
```

```
Final amount after 10% discount: Rs. 4500.0
Quantity updated to 5.
Invalid quantity.
Order ID: 1, Customer Name: Nimal, Item Price: Rs. 2500.0, Quantity: 5
Order ID: 2, Customer Name: Kamal, Item Price: Rs. 1500.0, Quantity: 3
```

Q3

```

1 public class StudentResult {
2     int studentId;
3     String studentName;
4     double mathMark;
5     double scienceMark;
6     double englishMark;
7
8     public StudentResult(int studentId, String studentName, double mathMark, double scienceMark, double englishMark) {
9         this.studentId = studentId;
10        this.studentName = studentName;
11        this.mathMark = mathMark;
12        this.scienceMark = scienceMark;
13        this.englishMark = englishMark;
14    }
15
16    public double calculateAverage() {
17        return (mathMark + scienceMark + englishMark) / 3;
18    }
19
20    public String getGrade() {
21        double average = calculateAverage();
22
23        if (average >= 75) {
24            return "A";
25        } else if (average >= 65) {
26            return "B";
27        } else if (average >= 50) {
28            return "C";
29        } else if (average >= 35) {
30            return "S";
31        } else {
32            return "F";
33        }
34    }
35
36    public void updateMark(String subject, double newMark) {
37        if (subject.equalsIgnoreCase("math")) {
38            mathMark = newMark;
39        } else if (subject.equalsIgnoreCase("science")) {
40            scienceMark = newMark;
41        } else if (subject.equalsIgnoreCase("english")) {
42            englishMark = newMark;
43        } else {
44            System.out.println("Invalid subject name.");
45        }
46    }
47
48    public String getResultSheet() {
49        return "Student ID: " + studentId +
50            "\nStudent Name: " + studentName +
51            "\nMath Mark: " + mathMark +
52            "\nScience Mark: " + scienceMark +
53            "\nEnglish Mark: " + englishMark +
54            "\nAverage: " + calculateAverage() +
55            "\nGrade: " + getGrade();
56    }
57
58    public static void main(String[] args) {
59        StudentResult student1 = new StudentResult(1, "Amal", 80, 75, 70);
60        StudentResult student2 = new StudentResult(2, "Saman", 60, 55, 50);
61
62        student1.updateMark("math", 90);
63        student1.updateMark("history", 70);
64
65        System.out.println(student1.getResultSheet());
66        System.out.println();
67
68        System.out.println(student2.getResultSheet());
69    }
70 }

```

```
Math Mark: 90.0  
Science Mark: 75.0  
English Mark: 70.0  
Average: 78.33333333333333  
Grade: A
```

```
Student ID: 2  
Student Name: Saman  
Math Mark: 60.0  
Science Mark: 55.0  
English Mark: 50.0  
Average: 55.0  
Grade: C
```

Q4

```

1  public class VehicleRental {
2      int vehicleId;
3      String vehicleType;
4      double ratePerDay;
5      double availableKm;
6      boolean isAvailable;
7
8      public VehicleRental() {
9          vehicleId = 0;
10         vehicleType = "Unknown";
11         ratePerDay = 0.0;
12         availableKm = 0.0;
13         isAvailable = true;
14     }
15
16     public VehicleRental(int vehicleId, String vehicleType, double ratePerDay, double availableKm, boolean isAvailable) {
17         this.vehicleId = vehicleId;
18         this.vehicleType = vehicleType;
19         this.ratePerDay = ratePerDay;
20         this.availableKm = availableKm;
21         this.isAvailable = isAvailable;
22     }
23
24     public String rentVehicle(int days, double estimatedKm) {
25         if (!isAvailable) {
26             return "Vehicle is not available.";
27         } else if (estimatedKm > availableKm) {
28             return "Estimated kilometers exceed available kilometers.";
29         } else {
30             isAvailable = false;
31             double totalCost = ratePerDay * days;
32             return "Vehicle rented successfully. Total cost: Rs. " + totalCost;
33         }
34     }
35
36     public void returnVehicle(double kmUsed) {
37         availableKm = availableKm - kmUsed;
38         isAvailable = true;
39         System.out.println("Vehicle returned. Available Km: " + availableKm + ", Available: " + isAvailable);
40     }
41
42     public String getVehicleInfo() {
43         return "Vehicle ID: " + vehicleId +
44             ", Vehicle Type: " + vehicleType +
45             ", Rate Per Day: Rs. " + ratePerDay +
46             ", Available Km: " + availableKm +
47             ", Is Available: " + isAvailable;
48     }
49
50     public static void main(String[] args) {
51         VehicleRental vehicle1 = new VehicleRental();
52         VehicleRental vehicle2 = new VehicleRental(101, "Car", 6000.0, 500.0, true);
53
54         System.out.println(vehicle2.getVehicleInfo());
55
56         System.out.println(vehicle2.rentVehicle(3, 200));
57         System.out.println(vehicle2.getVehicleInfo());
58
59         System.out.println(vehicle2.rentVehicle(2, 100));
60
61         vehicle2.returnVehicle(150);
62         System.out.println(vehicle2.getVehicleInfo());
63
64         System.out.println(vehicle2.rentVehicle(2, 100));
65         System.out.println(vehicle2.getVehicleInfo());
66     }
67 }

```

```
Vehicle ID: 101, Vehicle Type: Car, Rate Per Day: Rs. 6000.0, Available Km: 500.0, Is Available: true
Vehicle rented successfully. Total cost: Rs. 18000.0
Vehicle ID: 101, Vehicle Type: Car, Rate Per Day: Rs. 6000.0, Available Km: 500.0, Is Available: false
Vehicle is not available.
Vehicle returned. Available Km: 350.0, Available: true
Vehicle ID: 101, Vehicle Type: Car, Rate Per Day: Rs. 6000.0, Available Km: 350.0, Is Available: true
Vehicle rented successfully. Total cost: Rs. 12000.0
Vehicle ID: 101, Vehicle Type: Car, Rate Per Day: Rs. 6000.0, Available Km: 350.0, Is Available: false
PS C:\NCPM1> .\Car\CarAssignment1.ps
```

Q5


```

1  public class LibraryBook {
2      int bookId;
3      String title;
4      String author;
5      int totalCopies;
6      int borrowedCopies;
7
8      public LibraryBook() {
9          bookId = 0;
10         title = "Unknown";
11         author = "Unknown";
12         totalCopies = 0;
13         borrowedCopies = 0;
14     }
15
16     public LibraryBook(int bookId, String title, String author, int totalCopies) {
17         this.bookId = bookId;
18         this.title = title;
19         this.author = author;
20         this.totalCopies = totalCopies;
21         this.borrowedCopies = 0;
22     }
23
24     public String borrowBook(int count) {
25         int available = totalCopies - borrowedCopies;
26
27         if (count <= available) {
28             borrowedCopies = borrowedCopies + count;
29             available = totalCopies - borrowedCopies;
30             return "Successfully borrowed " + count + " copy/copies. Remaining: " + available;
31         } else {
32             return "Only " + available + " copy/copies available.";
33         }
34     }
35
36     public void returnBook(int count) {
37         if (count <= borrowedCopies) {
38             borrowedCopies = borrowedCopies - count;
39             System.out.println(count + " copy/copies returned successfully.");
40         } else {
41             System.out.println("Return count exceeds borrowed copies.");
42         }
43     }
44
45     public int getAvailableCopies() {
46         return totalCopies - borrowedCopies;
47     }
48
49     public String getBookInfo(String shelf) {
50         return "Book ID: " + bookId +
51             ", Title: " + title +
52             ", Author: " + author +
53             ", Total Copies: " + totalCopies +
54             ", Borrowed Copies: " + borrowedCopies +
55             ", Shelf: " + shelf;
56     }
57
58     public static void main(String[] args) {
59         LibraryBook book2 = new LibraryBook(101, "Java Programming", "James Gosling", 5);
60
61         System.out.println(book2.getBookInfo("A1"));
62
63         System.out.println(book2.borrowBook(3));
64         System.out.println(book2.getBookInfo("A1"));
65
66         System.out.println(book2.borrowBook(4));
67
68         book2.returnBook(2);
69         System.out.println(book2.getBookInfo("A1"));
70
71         System.out.println(book2.borrowBook(2));
72         System.out.println(book2.getBookInfo("A1"));
73     }
74 }

```

```
Book ID: 101, Title: Java Programming, Author: James Gosling, Total Copies: 5, Borrowed Copies: 0, Shelf: A1
Successfully borrowed 3 copy/copies. Remaining: 2
Book ID: 101, Title: Java Programming, Author: James Gosling, Total Copies: 5, Borrowed Copies: 3, Shelf: A1
Only 2 copy/copies available.
2 copy/copies returned successfully.
Book ID: 101, Title: Java Programming, Author: James Gosling, Total Copies: 5, Borrowed Copies: 1, Shelf: A1
Successfully borrowed 2 copy/copies. Remaining: 2
Book ID: 101, Title: Java Programming, Author: James Gosling, Total Copies: 5, Borrowed Copies: 3, Shelf: A1
```